Two distinct causal beliefs organizing intuitions about mind, action, and body

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Abstract

Prior research reveals that people tend to hold two conflicting intuitions. First, people think other people have ethereal minds, which are a distinct kind of thing, and even exist independently, from physical actions and living bodies. But, people also expect mental states to routinely interact with physical actions and bodily events, for example by reasoning about mental states as the causes of physical actions. What allows human minds to maintain these opposing beliefs simultaneously? In this paper, we propose that these two beliefs reflect two kinds of causal knowledge: one that organizes people's capacities in terms of *causal origins* (e.g. a mind that gives the capacity to remember and think; a physiological body that enables being tired and hungry), and a second that organizes them in terms of the *causal chains* connecting specific capacities to each other (e.g. that remembering something can cause someone to feel hungry). We tested this proposal in 6 pre-registered experiments in US adults (total N = 670). In Experiment 1, we measured people's beliefs about the similarity and causal connections between 15 mental events, actions, and bodily events, and replicated the observation that people think of these 3 domains as distinct yet connected. In 5 further experiments (Experiments 2-6), we explored the causal content they articulate by investigating their cognitive functions. Causal origins were used to make inductive inferences, plan interventions, and reason about biological bases of behavior, and causal chains were used to explain, predict and intervene on other people's behaviors. These results bridge prior research on latent dimensions of mind (1), theory of mind (2), and mind-body dualism (3, 4), providing a unified account of three distinct and interacting branches of social knowledge: knowledge of other people's minds, actions, and physiological bodies.

Significance statement

One of humans' most deeply held beliefs is that the mind is a distinct entity from the body. Here we show that US adults also maintain a separate, second set of beliefs about how specific events of the mind (e.g. seeing something) can lead their bodies to act in specific ways (e.g. reaching for something), and to experience particular bodily sensations (e.g. feeling hungry). People used both intuitions -- one about causal origins, and the second about causal sequences -- to guide their explanations, predictions, and decisions when interacting with others. Thus commonsense understanding of other people is multidisciplinary, jointly drawing upon knowledge of their status as mental beings, physical actors, and living things.

Main Text

Since antiquity, philosophers and scientists have puzzled over the structure and function of our brains, minds, and bodies, as well as how they relate to each other (5–7). Although these questions are active areas of academic research, they also animate our everyday reasoning. Prior research has explored people's folk understanding of other people, who are simultaneously mental beings with capacities to sense and think, physical bodies that carry out actions in the world, and living things that need food and nourishment to survive and whose bodies can be injured and healed. This research has revealed two opposing findings. First, people hold the intuition that physical and living bodies are a distinct kind of thing from immaterial mental states (1, 8, 9). At the same time, people think about bodies as causally linked to mental states, as in theory of mind (10–12). What makes it possible for human minds to hold these two, apparently incommensurate, beliefs simultaneously? Here we propose that this is possible because these beliefs express two sorts of causal knowledge, which are functionally useful for solving different sorts of reasoning problems. One representation describes the common *causal origins* of people's capacities, exemplified by variables like "the mind" (which causes perceiving and thinking), "the mechanical body" (which causes moving and reaching) and "the physiological body" (illness and hunger). A second representation describes the *causal chains,* or sequences of cause-and-effect relationships between specific events (e.g. perceiving causing thinking, and thinking causing moving).

There is a long history of studying the two contrasting intuitions—mind and body as distinct, and mind and body as connected—across the cognitive sciences. First, a vast literature shows that people represent the mind, the physical body, and the physiological body using distinct cognitive and neural resources. From childhood, people understand psychological, physical, and biological phenomena via distinct explanatory frameworks, leading to the proposal that people hold intuitive theories for each of these three domains (13–17). Intuitive theories of other people's minds posit mental states like beliefs, costs, and rewards as the causes of their actions (2, 11, 18–20). Intuitive theories of the physical world posit variables like mass, force, and objecthood, and principles that govern physical interactions (21–24). And intuitive theories of the biological world posit hidden variables like an innate potential or a hidden 'life force' that govern processes like life and death, injury and healing, and growth and reproduction (25–28). Even in the absence of an explicit reasoning task, children and adults from different cultures draw distinctions between events of the physiological body (e.g. feeling pain) and events of the mind (e.g. thinking about things) (1, 29–32). These distinctions are supported by evidence from neuroimaging: distinct brain regions respond preferentially to the mental vs bodily states of other people (e.g. beliefs and desires vs hunger and pain) (33, 34). At the most extreme, people have been shown to think about the mind as existing independently of the body (3, 4, 9), such that a person's body can be destroyed in death while their mind persists, or cloned without also copying over their mental states.

At the same time, prior work has also documented people's intuitions about connections between minds and bodies. People represent the mind as capable of causally interacting with both agents' own physical and living bodies. Most prominently, a theory of mind is not merely a theory of mental states, but rather how mental states cause other people's overt behaviors (12, 35–38). The connection between psychological states and physical action is available to young children, who appreciate that wanting and believing something can lead to acting (39–41), and even infants, who interpret the physical actions and the changes they effect in terms of mental states like knowledge, preference, and intent (42–45). One interpretation of these findings is that in order for people to understand other people's actions and minds at all, we must simultaneously represent other people as minds and physical bodies, acting in a physical world (46, 47). By adulthood, people easily conceive of mental states as leading to bodily states (48), and young children can learn from just a few pieces of evidence that a mental state (e.g. thinking about something scary) can lead to a bodily experience (e.g. a stomachache) (49). Thus, although people make a strong distinction between people's minds and their physical bodies, by adulthood, people may have strong and systematic intuitions about how minds affect bodies and vice versa.

Why do people simultaneously believe that minds are distinct from bodies, and yet connected to them? In philosophy, scholars have approached this contradiction by debating in what sense minds and bodies are fundamentally distinct, and whether the apparent causal link between the mental and physical worlds are true or epiphenomenal (see also (50)). From the perspective of cognitive science, however, one intriguing answer could be that maintaining this apparent contradiction has functional utility. People's experiences unfold over temporal sequences that regularly cross ontological boundaries; people's physiological states affect their mental states (e.g. feeling sick can lower someone's mental sharpness; or feeling hungry can make someone desire to eat), and actions (e.g. making a doctor's appointment or; walking into a restaurant) (51). Although none of these causal relationships are directly observable, navigating our social worlds involves a kind of problem-solving which would be supported by an understanding of sequences of cause-and-effect relationships between events (*causal chains*), that freely cross domain boundaries. Another kind of socio-cognitive problem might require us to go beyond the observable in a different way, to consider the underlying causes that give people the mental, physical and, and biological capacities that they have (*causal origins*), that make clear delineations between the mind, the mechanical body, and the living body. These two sorts of causal relationships (common causes vs causal chains; $A \leftarrow C \rightarrow B$ vs $A \rightarrow C \rightarrow B$) have been well studied in formal theories of causation as well as causal thinking and learning (e.g. (52–57)). Thus in the current work, we explore how these two kinds of causal representations, *causal chains* and *causal origins*, help us make sense of human social cognition in complementary ways.

Current research

In the current research, we combine the data- and theory-driven approaches developed in prior work to address this question, by shedding light on two types of causal knowledge of other people as physical actors, minds, and living systems (**Fig 1**).

The first goal of the current work is to use common methods to characterize and compare how people explicitly (1) organize and (2) think causally about mental events, actions, and physiological events, in a bottom-up manner. In Experiment 1, we measured people's intuitions about how actions, mental events, and bodily events "go together", as well as their beliefs about how strongly each pair of these events are causally connected. We found that people's judgments about which events go together reflected the latent dimensions discovered in prior literature, but these dimensions failed to describe people's beliefs about the causal relationships between these events, which freely crossed boundaries of mind, action, and body, in systematic and asymmetrical ways.

After confirming that people simultaneously organize events of the mind, actions, and events of the body as distinct and connected, we then tested the proposal that the two representational geometries we measured in Experiment 1 capture distinct frameworks that structure our causal understanding of other people. In Experiments 2-6, we studied the nature and functions of these two representations in explaining, predicting, and planning actions in the social world. Overall, we found that people prioritized causal chains to explain, predict, and intervene on specific actions, mental and bodily events, and prioritized causal origins to make inductive inferences about their common causes.

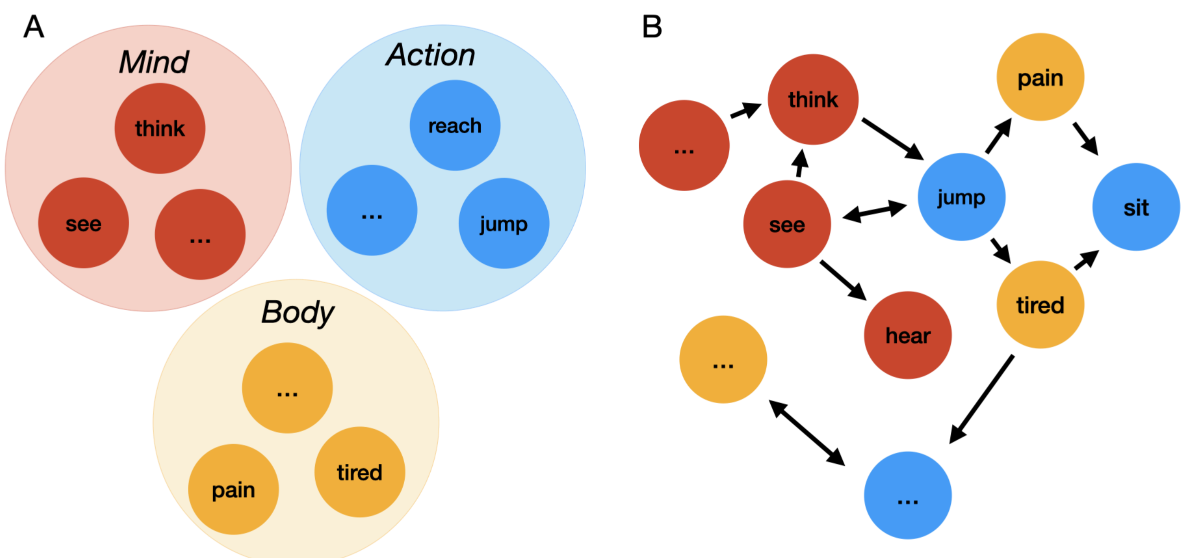

Fig 1. Two types of causal knowledge people have about other people's minds, actions, and bodies. (A) People stipulate boundaries between the psychological mind, physical body, and living body, and use these boundaries to organize events in terms of their *causal origins*. (B) People represent the asymmetric ways that mental states, actions, and bodily states connect via *causal chains*, both to themselves and to each other, in a manner that is neutral to the boundaries from (A).

Experiment 1: Two distinct representational geometries organize intuitions about minds, bodies, and actions

In Experiment 1, pre-registered at OSF, we measured how US participants (N = 101) organize 15 mental events, actions, and bodily events from prior work (1, 8, 29). The mental events included perceptual and cognitive events, like seeing and thinking; actions included both object- and non object-directed actions, like reaching for something and jumping; and bodily experiences included phasic and drawn out events, like feeling pain and getting tired (see **Fig 2**). Each participant engaged in two tasks presented in a random order: a Freesort task, where they placed the 15 items on a circular canvas, grouping similar items together, and a Causal Pairs task, where they judged whether experiencing one event could make someone experience a second on a sliding scale ("definitely not" to "definitely yes"), for all pairs of unique items. We then used representational similarity analysis (RSA) (58) to characterize the representational geometry of judgments produced by each task. Responses from the Freesort Task were transformed into Euclidean distances between pairs of items (*freesort distances*) and responses from the Causal Pairs task yielded *causal pairs distances*. **Fig 2** shows the representational dissimilarity matrices (RDMs) from each task which shows the distance between each pair of items, ranging from 0 (spatially close, or definitely not causally related) to 1 (far, or definitely causally related).

We pre-registered the hypothesis that the spatial clustering of these 15 items would fall into 3 a priori categories, as in prior work (1), but that the causal judgments about them would substantially differ from these categories. Because people's responses during each task were similar regardless of task order, we did not consider task order as a variable of interest in subsequent analyses (see SI for details).

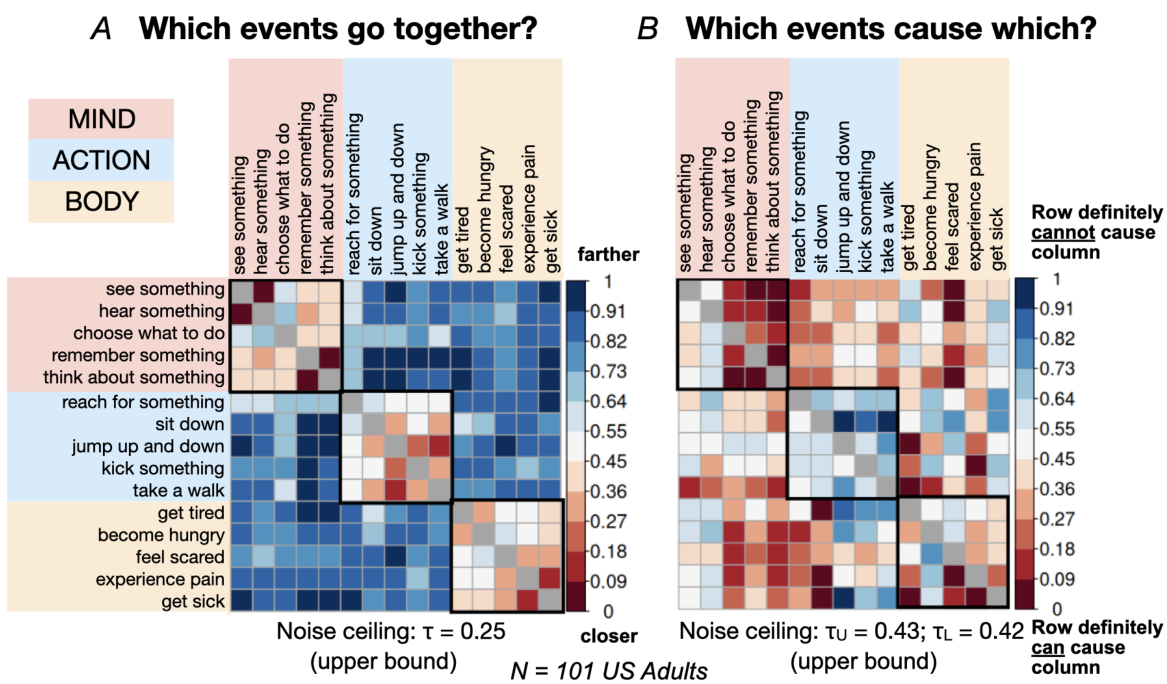

Figure 2. Results for Experiment 1. (A) Group-averaged representational dissimilarity matrix (RDM) for responses in the Freesort Task, achieved by computing the normalized Euclidean distance between the sorted items for each participant, and averaging across participants. (B) Group-averaged RDM for the Causal Pairs Task, achieved by computing the normalised causal ratings between all pairs of items for each participant, and averaging across participants. Closer items (Freesort Task) and more causally connected items (Causal Pairs Task) are plotted in

warmer colors, and farther items (Freesort Task) and less causally connected items (Causal Pairs Task) are plotted in cooler colors. Because the RDM from the Freesort Task is by design symmetrical, its upper bound noise ceiling reflects just the upper diagonal; by contrast we report the upper bound noise ceiling for the upper and the lower diagonal separately (τ_U, τ_L) for the Causal Pairs Task. Items are color-coded by a priori domain categories (mind, action, and body).

In confirmatory analyses, we found that in the Freesort task, participants grouped the 15 items into three distinct clusters of mind, action, and body. A linear mixed-effects model (Kendall's τ ~ model_RDM + (1 | subject id)) showed that appealing to just the three categories described the distances from this task (τ = 0.27 [0.24, 0.30], p<.001), outperforming alternative models that make broader distinctions between the events (2 groups, clustering actions and bodily events together, apart from mental events; difference in similarity relative to the best model, Δτ= −0.10 [−0.13, −0.07], *p*<.001), as well as those that make finer-grained distinctions (two subcategories within each of the 3 groups, Δτ = –0.08 [−0.1, −0.05], *p*<.001), and a phrase embeddings model that used the semantic similarity between 512-dimensional vectors as a distance metric (Δτ = –0.12 [–0.15, −0.09], *p*<.001) (see **SI** for details). The best model performed comparably to the noise ceiling for this task (τ = 0.25), suggesting that it explained virtually all the explainable variance in participants' responses.

By contrast, when asked about the pairwise causal relationships between mental events, actions, and bodily events, people's judgments freely crossed the 3 boundaries (**Fig 2B**). The best RDM model for the Freesort Task data provided a poorer fit for the Causal Pairs Task data (Δτ = −1.42 [−1.61, −1.22], *p*<.001). In fact, all models were negatively associated with people's responses in the Causal Pairs Task, indicating a systematic mismatch with the data (τ = −0.03 [−0.05, −0.01], *p*<.001 for mind, action and physiology baseline model; and differences relative to baseline of Δτ = −0.03 [−0.05, −0.02] to −0.06 [−0.07, −0.04], all *ps* < .001; see Supplementary Figure S1). For example, although sitting and jumping were sorted near to each other (mean freesort distance = 0.28 [0.22, 0.33]), people judged that sitting cannot cause jumping to happen (mean causal pairs distance = 0.82 [0.77, 0.87]) and vice versa (0.53 [0.46, 0.60]). Yet, people's responses were not merely noisy (in fact, the noise ceiling for the Causal Pairs task was higher than for the Freesort task). Instead, we observed systematic patterns of results that re-capitulated prior findings on theory of mind. For example, participants reported that mental events are more likely to cause actions (0.31 [0.30, 0.32]) than the other way around (0.38 [0.36, 0.39]), (β = 0.22 [0.17, 0.26], p<.001), and that perception (seeing and hearing) was more likely to influence cognition (deciding, thinking, remembering) (0.15 [0.13, 0.17]) than vice versa (0.48 [0.44, 0.51]), (β = 1.04 [0.96, 1.13], p<.001) (39, 42). This bottom-up approach also revealed findings that went beyond prior work. For example, people believed that mental states were causally promiscuous; that is, more likely to both cause other states and themselves (0.31 [0.31, 0.32]), compared to actions (0.44 [0.43, 0.45]), (β = 0.41 [0.37, 0.43], p<.001) and physiological states (0.37 [0.36, 0.37]), (β = 0.17 [0.14, 0.20], p< .001). While actions were rated the least causally promiscuous overall, this differed across actions: for example, taking a walk was more likely to cause changes in the mind (0.22 [0.20, 0.24]) compared to the causal potential of other actions (0.34 [0.33, 0.34]), (β = 0.40 [0.32, 0.48], p<.001). Lastly, participants thought that bodily states could cause mental states (0.34 [0.32, 0.35]) more strongly than vice versa (0.36 [0.35, 0.37]), (β = 0.07 [0.02, 0.12], p <.001). Despite sorting bodily states apart from actions and mental states, participants demonstrated systematic beliefs about how they interact: for example, bodily states were more likely to influence cognitive (0.25 [0.23, 0.26]) than perceptual events (0.47 [0.46, 0.49]), (β = 0.73 [0.66, 0.80], *p*<.001) and actions (0.42 [0.40, 0.43]), (β = 0.54 [0.49, 0.60], p <.001). In sum, when we directly compared participants' clustering of actions, mental events, and physiological events, and their causal beliefs about which events can influence which, we observed two distinct representational geometries, that both recapitulated prior work on latent dimensions and intuitive theories, and contributed novel insights about their fine-grained relational structure and how they interact with each other.

Experiments 2-6: People use representations of causal origins and causal chains during everyday cognition

In Experiment 1, we found that when US adults are explicitly asked to do so, they organize mental events, bodily events, and actions using two distinct representational geometries. What is the nature of the representations in these two spaces, and what functional role do they play in cognition? We hypothesized that people's intuitions about which events go together, as well as their explicit causal beliefs about which events can cause which, reflect their intuitive theories of how other people's minds and bodies work. Under this account, people organize mental events, bodily events, and actions in two parallel ways: First, they posit the causes responsible for generating events of the mind, actions, and events of the body (*causal origins*) (26). Second, they represent the pairwise causal relationships between all of these events, capturing expectations about how events within and across domains interact (*causal chains*) (59).

In Experiments 2-6, we tested how causal origins and causal chains support functions like prediction, explanation, counterfactuals, and intervention (19, 60–63). Our general strategy was to (1) use the group-averaged judgments from Experiment 1 and an additional large pilot study (total N = 150) to construct trials that pit causal origins and causal chains against each other, and (2) embed these stimuli in tasks that we hypothesized would preferentially rely on one of these representations. For each target item in these studies, we selected a *causal chain (CC) item*, that was most strongly related to it via a causal chain relative to a shared causal origin, and a *causal origin (CO) item*, that was most strongly related via a shared causal origin relative to a causal chain. For example, the target item "jump up and down" was presented with the CC item "see something" (freesort distance 0.60, causal pairs distance 0.29) and the CO item "sit down" (freesort distance 0.20, causal pairs distance 0.83) (**Fig 3A**). Detailed information about trials for each experiment can be found in the SI.

Experiments 2-3: Using causal chains during explanations and counterfactual reasoning.

First we investigated whether causal beliefs that we explicitly measured in Experiment 1 are used for classic functions of intuitive causal reasoning, like explanation and counterfactuals (61, 63, 64). (**Fig 3B-C**). We hypothesized that people would preferentially rely on causal chains (e.g. that jumping up and down can make someone feel tired) to explain what happened, and to reason counterfactually about whether the same thing would have happened if the state of affairs were different.

In Experiment 2, participants were presented with causal chain (CC) items and causal origin (CO) items (15 trials each) as potential explanations for a target outcome (**Fig 3B**), and rated on a continuous slider how satisfactory the explanations were ("not at all satisfying" = 0, to "extremely satisfying" = 100). For example, participants considered to what extent "sitting down" (CO item) and "thinking about something" (CC item) were satisfying explanations for "taking a walk". Overall, people judged that CC items (M = 58.38 [55.76, 60.99]) are better explanations than CO items (M = 34.87 [32.25, 37.48]), (β = 0.72 [0.66, 0.78], $p<.001$).

In Experiment 3, participants were first told that a causal chain item or causal origin item (15 trials each) was followed by a target outcome, and then rated on a continuous slider how likely the target outcome would happen if the CC or CO item did not happen ("Definitely not" to "Definitely yes") (**Fig 3C**). People judged that CC items were more counterfactually relevant; specifically, blocking a CC item from happening would be more likely to prevent an outcome from happening (M = 34.01 [26.04, 31.98]), compared to blocking an CO item (M = 52.88 [50.84, 54.90]) (β = −0.62 [−0.68, −0.55], $p<.001$).

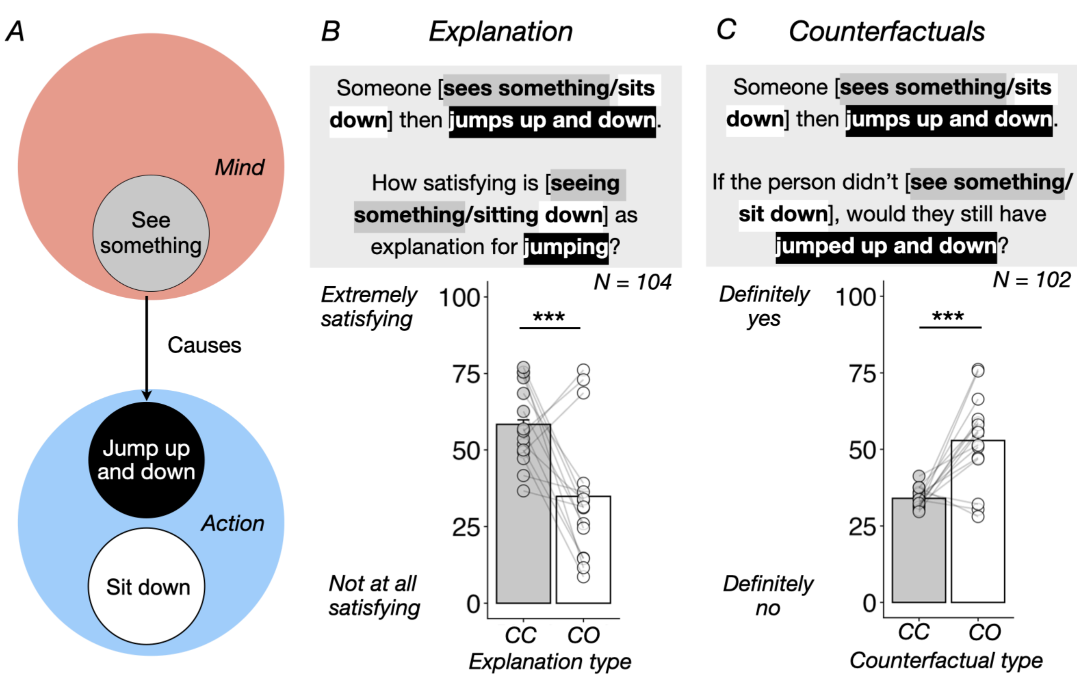

Fig 3: Method and results for Experiments 2 (Explanation) and 3 (Counterfactuals). (A) For each target item (black), we used the distances from Experiment 1 to choose a causal chain (CC) item (grey, i.e. the item sorted farthest to and rated most likely to cause the target), and a causal origin (CO) item (white, i.e. the item sorted closest to and rated least likely to cause the target). (B) In Experiment 2 participants rated how satisfying the CC and CO items are as an explanation for the target event. Results show CC items were rated as more satisfactory than

CO items. (C) In Experiment 3 participants rated how likely a target event is to occur if its preceding CC and CO event was blocked in a counterfactual scenario. The target item was rated as less likely to occur if the CC item was blocked in the counterfactual. Error bars indicate 95% confidence intervals, points indicate the average effect from a single trial (30 total per experiment), and pairs of points connect the chosen CC and CO item for each target. *** $p < .001$, two-tailed.

Experiments 4-6: Contrasting causal chains and causal origins for prediction, inference, and intervention

In Experiment 2-3 we showed that people prioritize causal chains for explanation and counterfactual reasoning, but when do people prioritize causal origins? Thus we designed 3 additional studies that randomly assigned participants to contrasting tasks that we hypothesized to recruit one representation more so than the other (**Fig 4A-C**). The dependent measure across these experiments was a binary forced choice between a causal chain (CC) or a causal origin (CO) item.

In Experiment 4, we compared inductive trait inference and prediction. Based on prior evidence that people use hidden essences to make inductive inferences about unseen properties (65), and that people use an understanding of cause-and-effect relationships to predict and explain (61), we hypothesized that people would prioritize causal origins for inductive inference and causal chains for prediction. Thus in Experiment 4, participants were randomized into either an Inductive Inference Task where they were asked to infer which capacity a novel creature would have given that it has a target capacity (15 trials), or a Prediction Task where they were shown a target event that happened to someone (15 trials) and asked what will happen next (**Fig 4A**). We predicted and found that people chose the CO item more often than chance during the Inductive Inference Task (Mean proportion = 0.56 [0.52, 0.61]), and less often than chance during the Prediction Task (0.18 [0.16, 0.22]; OR between conditions = 5.69 [4.33, 7.47], $p < .001$).

In Experiment 5, we were inspired by prior work suggesting that a truly causal belief not only allows the holder of the belief to explain and predict, but also guides plans to change states of the world (62, 66, 67). Participants were randomized into two tasks that both required them to plan an intervention, one of which made use of causal origins, and one which made use of causal chains. In the Magical Intervention Task, people were asked to imagine they were a wizard casting a spell to grant new creatures certain abilities. On each trial their goal was to grant a creature an ability (such as the ability to sit down) but happen to be out of that particular spell; instead, they were asked to choose between two other spells, one that relates to the target ability via shared causal origin (such as granting the ability to experience pain) and a second that relates to it via a causal chain (such as granting the ability to jump up and down) (**Fig 4B**). By contrast, in each trial in the Ordinary Intervention Task, participants were asked to make another person do something, and to choose an intervention that would likely to succeed (e.g. to make someone sit down, participants could either choose to make them experience pain or jump up and down). We predicted and found that people chose the CO item more often than chance during the Magical Intervention Task (0.56 [0.53, 0.60]), and less often than chance during the Ordinary Intervention Task (0.35 [0.31, 0.38]; OR between conditions = 2.56 [1.90, 3.44], $p<.001$). Thus people use representations of causal origins and causal chains alike to plan actions in order to achieve their goals, strongly suggesting that both reflect genuine causal beliefs.

In Experiment 6, we explored two further functions of causal origins and causal chains: reasoning about biological causes and reasoning through multiple links of a cause-and-effect sequence. Prior work suggests that people represent hidden essences of kinds, or placeholders for inductively rich properties characteristic to that kind (68–70), and are particularly drawn to explanations that support the existence of these placeholders (71). Thus in the Biological Reasoning Task,we tested whether people would preferentially use causal origins to guide reasoning about potential biological substrates of mental events, actions, and bodily events. People were told to pretend that they are scientists studying the biological basis of behaviors in new creatures, and that they have discovered a novel gene that controls two of their capacities (e.g. sitting down and jumping up and down). They were asked to infer which third capacity is controlled by the same novel gene: one linked strongly to the other two in terms of causal origins (e.g. taking a walk), or in terms of a causal chain (e.g. feeling pain). In the other condition of Experiment 6, we were inspired by the capacity for causal representations to link together otherwise seemingly independent events; for example, although sitting down is not likely to cause someone to jump up and down, it can cause someone to feel pain, which could cause them to jump up and down. In the Causal Triads Task, people were asked to choose one of two events that completes the sequence [X → [?] → Z], with the same stimuli and items as the Biological Reasoning Task (**Fig 4C**). We predicted that people would prioritize causal origins to make sense of the biological causes of behavior, and causal chains when linking up a sequence of causes and effects. Our findings supported these predictions: People were more likely than chance to choose the CO item in the Biological Reasoning Task (0.91 [0.87, 0.93]), and less likely than chance to choose the CO item in the Causal Chain Task (0.30 [0.24, 0.35]; OR between conditions = 23.2 [14.80, 36.20], $p<.001$).

In summary, when people were asked to predict what will happen next (Exp 4), choose which event would trigger a second event (Exp 5), and infer the missing middle event in a sequence(Exp 6), they were more likely to rely on causal chains than causal origins. By contrast, when people were asked to make a trait inference (Exp 4), choose which of two spells would indirectly grant a creature an novel capacity (Exp 5), and infer which missing capacity is controlled by the same biological substrate (Exp 6), they were more likely to rely on causal origins. Together with Experiments 2-3, we interpret these results as evidence that the two representational geometries we measured in Experiment 1 constitute two sets of causal beliefs about how other people work: one that describes the hidden essences that are responsible for generating their capacities and a second that describes how these capacities are causally related.

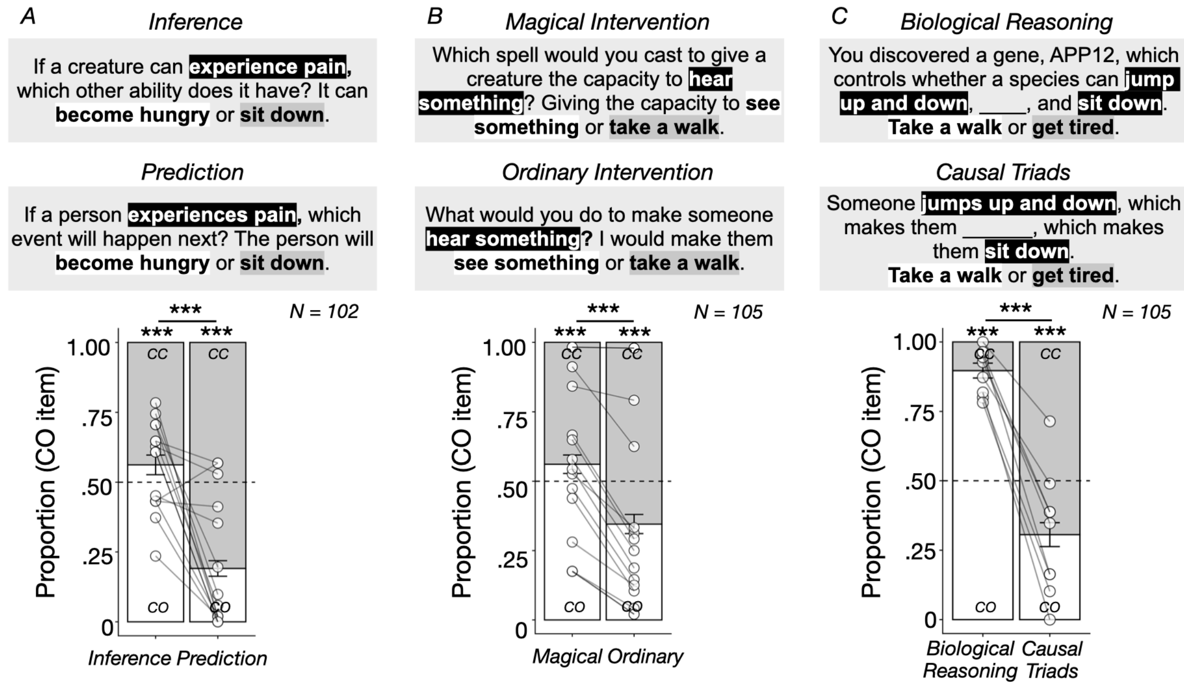

Fig 4: Method and results for (A) Experiment 4 (Inductive Inference vs Prediction), (B) Experiment 5 (Magical vs Ordinary Intervention), and (C) Experiment 6 (Biological Reasoning vs Causal Triads). Results showed that people used causal origins (CO) to make an inductive inference, plan an indirect intervention based on that inference, and identify the biological substrates of mental events, actions, and bodily events. By contrast, people relied on causal chains (CC) to make a prediction, plan a direct intervention, and complete a chain of causes and effects. Error bars indicate 95% confidence intervals, and pairs of points indicate averaged effects from the same stimuli presented across both conditions . *** $p < .001$, two-tailed.

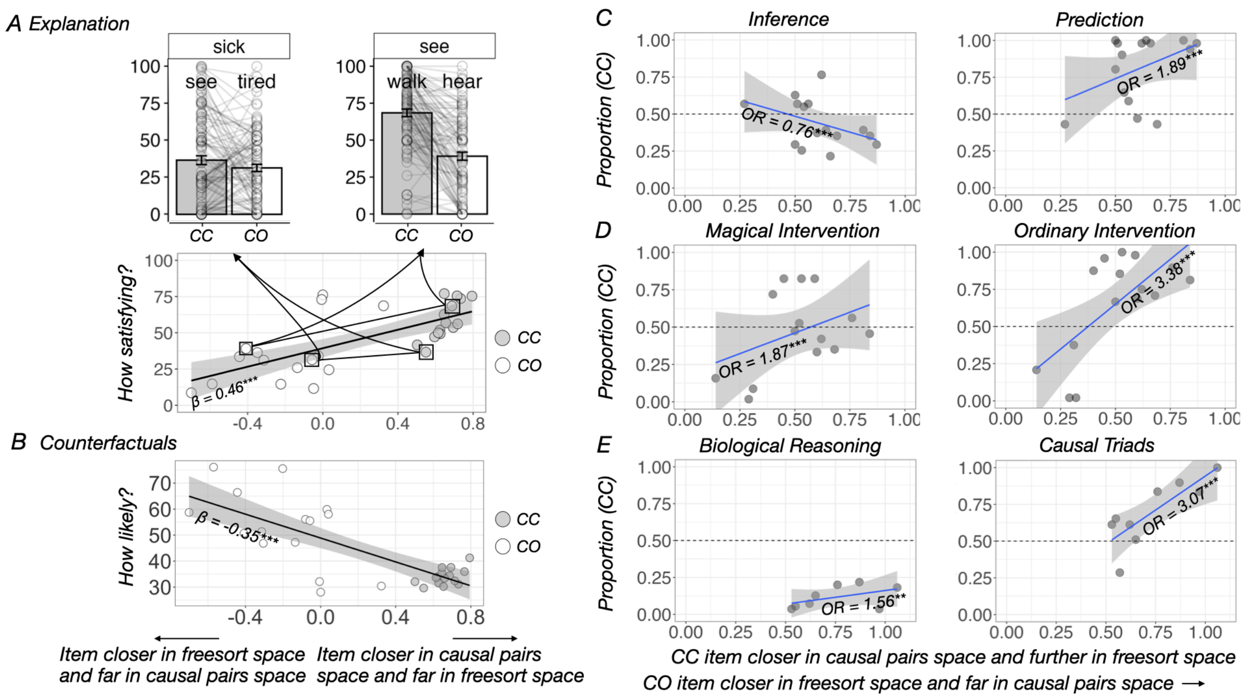

Fig 5: Using quantitative distances to predict trial by trial effect sizes. (A-B) In Experiments 2-3 (Explanation & Counterfactuals), the items providing the most satisfying explanations and the items that were most counterfactually relevant (i.e. the ones that would maximally block a target event from happening) were those most close by in causal pairs space and faraway in freesort space, in terms of distances from Experiment 1 targets. (C-E) In conditions in Experiment 4-6 hypothesized to rely on causal chains (Prediction, Ordinary Intervention & Causal Triads), the strongest effects appeared during trials where the CC or CO item was strongly related to the target via a causal chain and weakly related via a causal origin. Among the conditions in Experiment 4-6 hypothesized to rely on causal origins (Inference, Magical Intervention & Biological Reasoning, we found the strongest effects for pairs where the CO item was strongly related to the target in terms of causal origins and weakly related to it in terms of causal chains only in Experiment 4. This suggests that representations of causal chains influenced people's responses, but not strongly enough to override an overall preference for the causal origin item. ** $p < .01$, *** $p < .001$, two-tailed.

Predicting trial-by-trial effect sizes from distances

In the confirmatory analyses in Experiments 2-6, we found that people intuitively and flexibly prioritized causal origins and causal chains in different tasks. However, there is likely no 1:1 mapping between cognitive functions and the two representations (e.g. that explanations *always* will draw upon causal chain representations). This is demonstrated by Experiment 5, which found that subjects can use either causal chains or causal origins to design interventions.

Across all experiments, we observed substantial item-level variability in effect sizes. For example, in the Prediction Task in Experiment 4, people respond that jumping is much more likely to make someone get tired, than to make them take a walk (Mean proportion selecting tired = 0.94 [0.85, 0.99]) (OR = 13.61 [6.85, 20.37], $p<0.001$), whereas hearing is just as likely to make someone feel scared as to make them see something (mean proportion selecting feel scared = 0.47 [0.34, 0.61]) (OR = −0.12, [−0.67, 0.43], $p=.670$). What explains this variability? As an exploratory analysis, we asked whether variability in effect sizes is partially explained by variability in freesort and causal pairs distances measured in Experiment 1. If people are indeed using the spaces from Experiment 1 in Experiments 2-6, then we should observe the strongest effects in trials that maximize the hypothesized relevant distance (causal pairs or freesort, depending on the task).

We found that for all 5 tasks hypothesized to call upon causal chains, there was a robust relationship between trial-by-trial variability in item distances and people's behaviors (see Methods for details). In Experiments 2-3, the most satisfying explanations, and the most counterfactually relevant items, were those most close by in causal pairs space and far away in freesort space to the targets (Exp 2: $\beta = 0.46$ [0.43, 0.49], $p<.001$, Exp 3: $\beta = -0.35$ [−0.38, −0.32], $p<.001$). In Experiments 4-6, people most strongly chose the CC item over the CO items when the CC item was close in causal space and far away in freesort space, and when the distractor CO item was close in freesort space and far away in causal pairs space (Exp 4: $OR_{Prediction} = 1.89$ [1.55, 2.31], $p<.001$, Exp 5: $OR_{Ordinary\ Intervention} = 3.38$ [2.72, 4.19], $p<.001$, Exp 6: $OR_{Causal\ Triads} = 3.07$ [2.26, 4.17], $p<.001$).

The relationship between effect sizes and distances was less consistent for the 3 tasks hypothesized to rely on causal origins. Although people chose the CO item when making an inductive inference (Exp 4), designing an indirect intervention (Exp 5), and attributing biological causes (Exp 6), we only observed an orderly relationship between preference for the CO item and item distances in Experiment 4. Here, people were more likely to choose the CO item when making an inference about what other capacity a creature has when it was sorted closely to and not likely to cause the target (Exp 4: OR= 1.32 [1.13, 1.53], $p<.001$). By contrast, we did not observe this effect in Experiments 5-6. Here, although people chose the CO item overall, they were still swayed by causal chain distances, such that trials with the biggest effects had a small causal pairs distance to and a large freesort distance to the target (Exp 5: OR = 1.87 [1.59, 2.19], $p<.001$; Exp 6: OR = 1.56 [1.17, 2.08], $p<.001$). See **Figure 5**.

In summary, in addition to our pre-registered effects, we found systematic relationships between the incidental variability in item distances and effect sizes in 7 out of 8 conditions, 6 of which were predicted by which sort of cause they were hypothesized to recruit. This strongly suggests that people were calling upon the two representational spaces we measured in Experiment 1 to guide their performance on the tasks in Experiments 2-6. Together, these 6 experiments show that we represent and make use of two sorts of causal beliefs to maintain the boundaries between events of the mind and events of the body, and reason about how they causally interact.

Discussion

Prior research has demonstrated that we represent events of the mind, events of the body and actions as distinct and even independent (29, 31, 32), and at the same time, connected (72). What allows us to hold both sets of beliefs at the same time? In the current work, we tested the intriguing hypothesis that these two opposing intuitions are supported by two sorts of causal representations that help us navigate our social worlds: first, intuitions about the *causal origins* that give rise to the specific experiences people have and second, intuitions about which specific experiences can lead to others via *causal chains*.

In Experiment 1, we first confirmed that we can measure these two opposing beliefs using a common methodological framework. Using representational similarity analysis, we found a space that replicates the prior result that people think about certain events as “going together”, in a way that separates events of the mind, actions, and events of the body (3, 4, 29). We also found a second space which draws asymmetric causal connections within and across the three categories, and that freely crossed category boundaries, both at a broader scale (e.g. mental states tend to cause actions more than vice versa), and at a finer scale (e.g. jumping up and down tends to make others tired, but not sick).

In 5 further experiments (Experiments 2-6), we characterized the nature and function of these two representational spaces in social cognition. We hypothesized and found that people rely on causal chains when reasoning about the cause-and-effect relationships between pairs of specific events: when explaining, predicting, reasoning counterfactually about, intervening to bring about a specific state, and when making sense of how two otherwise unconnected events could be linked together into a coherent sequence. By contrast, we predicted and found that people rely on causal origins when reasoning inductively about hidden common causes that structure agents’ capacities into minds, actions, and bodies: when making inferences about, intervening on a novel creature’s capacities, and reasoning about the biological causes of their behavior. Not only did people prioritize causal chains vs causal origins for different tasks, but the extent to which they did so was also predictable from trial-by-trial differences in distances measured in Experiment 1, with strongest evidence in experiments targeting causal chains. From these findings, we infer that causal chains and causal origins support two parallel systems of causal beliefs articulating our intuitive theories of other people. They could support distinct modes of construal (59, 68) for reasoning causally about other agents: first, a mechanistic construal that provides an understanding of how people’s minds and bodies function together, and an essentialist construal that defines what it means to have a mind, an acting body, and a living body.

Our work contributes to the literature on social cognition in several other ways. First, prior work on the broad architecture, content, and functions of theory of mind leave open detailed descriptions of people’s causal beliefs. For example, studies of theory of mind have characterized how perceiving leads to knowing (40, 42, 73), how desire and belief jointly lead to action (39), and how agents plan over the costs of actions and the rewards they deliver (37, 38, 43). Yet, a fuller picture of the causal structure of our intuitive theories of mind should describe other causal relationships, for example pointing from actions to mental states (e.g. turning a corner can lead one to learn something new). Second, prior work on intuitive theories of minds and living things leaves open how they interact with each other. By taking a bottom-up approach, we discovered several new phenomena addressing both of these open questions: beliefs about the mental effects of taking actions, the mental causes of bodily events, and the bodily causes of mental events.

Our results also address open questions from prior work about the nature of these two kinds of commonsense beliefs. When adults and children organize social phenomena in terms of dimensions like “mind”, “heart”, and “body” (1), what do these dimensions reflect, and do these dimensions impose domain-specific constraints on the expectations connecting events into sequences of causes and effects? Ahead of our results, one possibility was that the dimensions could reflect associations between features, much like the high-dimensional spaces in which features are represented in modern artificial intelligence systems (74). Another possibility was that these dimensions could reflect a folk theory which functions in helping us navigate our social worlds (52, 68, 75–77). Grey et al (31) tested this by showing that given two dimensions of mind (agency, which most closely corresponds to “mind” in this paper, and experience, which most closely corresponds to “body”), people held agents with high agency more responsible for their actions, and agents with high experience as more deserving of protection against harm. Building on these results, our representational similarity results from Experiment 1 speak against the account that clusters of “mind”, “action”, and “body” reflect associations (78), and that people treat bodily events (actions and physiological states) under the same umbrella (8). Moreover, adults do not expect causal origins to govern how the cause and effect relationships work between specific mental events, actions, and bodily events. Further results from Experiment 2-6 suggest that these 3 dimensions reflect beliefs about the causal origins of agent’s experiences.

At the same time, there are open questions about how our minds represent causal origins and causal chains. For causal origins, one possibility is that people represent an explicit variable or process that is responsible for generating all the events in the variable’s domain, akin to essences for biological and social kinds (79, 80). Support for this view comes from the finding that people used the three clusters from Experiment 1 to infer biological causes. Yet, this paper does not establish that “mind”, “action” and “body” are the right level of description, because our findings are constrained by the space of events we chose to study. For example, it is possible that there is a finer-grained structure within each category that would be discoverable with more items. For causal chains, one open question is how people generated their intuitions that some events were definitely able to (or unable to) cause others, if in principle it is possible to imagine a situation wherein any pair of two events are causally connected. One possibility is that people answered this question by sampling specific scenarios instantiating the causal relation under discussion (e.g. “seeing something” -→ “feeling hungry”; “feeling sick” → “hearing something”). Under this hypothesis responses from this task reflect how easy it was for people to generate these samples, or how many samples people could generate.

Limitations

The current work has limitations. First, the structure and content of our intuitive theories about other people is more expansive and complex than the 15 events we studied in this paper. We focused on mental states, actions, and bodily states in order to better situate our results in prior literature on theory of mind, and the mind-body distinction. Second, these results only speak to social cognitive knowledge in US adults. Cross-cultural similarities and differences in social cognitive knowledge remains an area of active investigation (1, 47, 81–84). However, our methodological approach could be productively applied to study other topics in social cognition, including emotion[1], and moral reasoning, including in other cultures. For example, this approach could be used to shed light on the hypothesis that the human mind has two separate systems for social cognition, one that understands agents as instrumental actors and another that casts them as social beings (85), and test it against alternatives (e.g. that a social being is a special kind of agent).

Implications for cognitive development

How does our knowledge about the mind and body, as both distinct and connected entities, develop? One possibility is that children begin with an early-emerging ontological commitment that the physical and physiological body, and the immaterial mind, are distinct, and have to learn how they are connected (3). This account predicts that children begin to work out cross-domain connections by accumulating evidence about themselves and other people. Indeed, young children can learn about a mental cause of a bodily event with a few pieces of observational evidence (49). Yet, even children's most rudimentary understanding of agents includes the intuition that physical actions are guided by goals and intentions (86–88). Therefore, a second possibility is that children represent the mind and the body as causally connected entities in a limited and abstract way, which supports subsequent causal learning.

Conclusion

Over the course of childhood, people develop commonsense understandings of the psychological, physical, and biological world. Here we show that during social cognition, people flexibly use three domains of understanding to explain, predict, and intervene on other agents, who are minds, objects, and living things all at the same time. We propose that this flexibility comes from two parallel intuitive causal theories, which allow us to maintain the divide between agents' status as psychological, physical, and biological entities, and to reason about how they interconnect. More broadly, this work shows that fully understanding the nature and origins of social cognition will require a careful study of connections between multiple branches of commonsense knowledge, including those beyond the purview of the mind.

Materials and Methods

Participants

Participants in all studies were recruited via Connect CloudResearch and provided informed consent to participate. Study procedures were approved by the Institutional Review Board at Johns Hopkins University. Participants were paid $15 per hour for their time, and 0-5

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[1] Interestingly "feeling scared" was sorted as an event of the body in this study, and fell within the "body" dimension in Weisman et al. (1, 29).

participants were excluded per study for failing attention checks. See SI for study-specific demographic information about participants and exclusion rates.

Methods, materials and procedure

Open science practices. All experiment scripts and stimuli, data, analysis scripts are available at https://github.com/liulaboratory-experiments/mind-action-body. All methods and analyses were pre-registered on the Open Science Framework (Experiment 1: https://osf.io/xp8fu/overview; Experiment 2: https://osf.io/yahk7/overview; Experiment 3: https://osf.io/4mn2t/overview; Experiment 4: https://osf.io/nzj6y/overview, Experiment 5: https://osf.io/46sa5/files/bxnfh, Experiment 6:https://osf.io/7qcb3/overview).We disclose the following deviations from our pre-registration: In Experiments 4 and 5, we updated the pre-registration after spotting a change in wording of the instruction text comparing the main experiment to the original pilot; details of this update are available on OSF.

Experiment 1. During the Freesort Task, participants saw a circular sorting canvas with the 15 items placed in random positions outside the canvas. Participants were asked to take as long as they needed to drag and drop items in order to group them inside the canvas. Participants were not told about the three a priori categories.

In the Causal Task, participants saw the same 15 items presented in pairs, and reported whether experiencing one event could make someone experience the other. Participants gave responses for 210 pairs of the 15 items: this included every item paired with every other item in both orders (A causing B and vice versa), but excluded items paired with themselves. Trials were presented in groups of 15, either organized by the potential cause (e.g. Can jumping up and down make someone [...]?), or by the potential effect (Can [...] make someone jump up and down?), randomized across participants. For all participants, the particular order of items within a group, and the order of groups, were presented in a random order.

Stimulus selection and design, experiments 2-6. We used data from the Freesort and Causal Pairs Tasks for Experiment 1 and a large pilot study (total N = 152) to compute the normalized average distances between each pair of items for each task. Then we used these quantitative distances to construct sets of trials for each experiment consisting of a target item, a causal chain (CC) item, and causal origin (CO) item. For further details of how distances were used to select the CC and CO items, see SI.

For all experiments, trials were presented in a random order. In Experiments 2-3, participants saw target items paired with both an CC item and CO item, in separate trials. In Experiments 4-6, participants were randomly assigned to one of two conditions, but people in both conditions saw the same target items with a two-alternative forced choice between a CC item and CO item. For the full set of items for each experiment, see SI.

Statistical Analyses

All reported p-values in the paper are two-tailed, unless otherwise noted, and all bracketed values indicate 95% confidence intervals.

Confirmatory analyses, experiment 1. To calculate a noise ceiling of the data from each task of Experiment 1, we computed Kendall's τ between each participant's RDM and the average of all the other participants' RDMs, iterating through participants, and averaging across folds. We did this for the lower off-diagonial for the Freesort Task, and for the upper and lower off-diagonal for the Causal Pairs Task separately. People's empirical RDMs were compared to 4 theoretical RDMs (see SI): (1) *Psychological and physical* Inspired by the hypothesis that people draw a clear boundary between events of the physical world and events of the mental world (8) this model organizes events according to two categories: the mind (5 items) and the body (the remaining 10 items). (2) *Mind, Action and Body*: Organizes the items according to the 3 clusters: the mind, physical actions of the body, and physiological/biological bodily processes. (3) *Fine-grained Mind, Action, and Body:* This model makes one further distinction within each latent category proposed in (2) (i.e. perceptual vs cognitive events; object directed vs non-object directed actions; drawn out vs phasic bodily changes). (4) *Phrase embeddings:* This model uses a Universal Sentence Encoder to represent the 15 events as high dimensional vectors, derived from large scale corpora using machine learning techniques (78)

For each of these four models, for each participant, we computed Kendall's τ between that person's RDM and the theoretical RDM expressed by the model. For RDMs from the Freesort Task, we took the values from one half of the off-diagonal values, since the responses in this task were necessarily symmetrical; for RDMs from the Causal Pairs Task, we took the values from both halves of the off-diagonal matrix, since people gave ratings about both whether one item could cause a second, and vice versa. We pre-registered two main predictions and analyses. To test which of the models best explained the data from the Freesort Task, we compared the distribution of Kendall's τ across the four different models using a linear mixed effects model, with our preferred theoretical RDM (Mind, Action, and Body) as the baseline (`τ ~ model_RDM + (1 | subject id)`). To test whether the best model that accounts for the data from the Freesort Task provides a poorer fit to data from the Causal Pairs Task, we compared the fit of Mind, Action, and Body model to the empirical responses from the two tasks, using the Freesort Task as the baseline (`τ ~ task + (1 | subject id)`). For more details, see SI.

Confirmatory analyses, experiments 2-6. For Experiments 2-3, we modeled the continuous rating for each participant for each trial (satisfaction ratings or counterfactual likelihoods) as a function of item type (CC vs CO) in a linear mixed effects model including a random intercept for participant ID `lmer(response ~ trial_type + (1|subject id))`.For Experiments 4-6, we modeled the binary choices between the CC and CO item for each participant for each trial as a function of task condition in a logistic regression model, including a random intercept for participant ID: `glmer(response ~ condition + (1|subject id), family = binomial(link = 'logit'))`.

For all experiments, model quality assurance was conducted through inspection of the outputs from the `performance::check_model()` function.

Exploratory distance analyses, experiments 2-6. For each experiment, we tested whether the variability in item joint distances across trials significantly predicted people's behaviors. For Experiments 2-3, which paired one item (either CC or CO) with each target, this joint distance expressed how strongly people in Experiment 1 thought that the item could cause the target, relative to how closely they sorted the item relative to the target (for each pair, joint distance = freesort distance - causal pairs distance). For Experiments 4-6, which paired two items (one CC, one CO) as a 2-alternative forced choice with each target, this joint distance expressed how distinct the CC and CO item were, considering both freesort and causal pairs distances, from the target item (for each triad, joint distance = (freesort distance - causal pairs distance for CC item) - (freesort distance - causal pairs distance for the CO item)).

Pooling across all trials from Experiments and 3, and restricting the analysis to each condition of Experiments 4-6, we related these distance scores to how much one item was preferred over the other for each trial (`response ~ distance + (1|subject id)`).

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